

This is due by August 15.

Problem 1. Let $f(x) \in \mathbb{Z}[x]$ be primitive. Show $f(x)$ is irreducible iff $f(x)$ can't be factored into a product of two polynomial with strictly smaller degrees.

Problem 2. Determine if the following polynomials is irreducible in $\mathbb{Z}[x]$ and explain why.

1. $2x - 1$
2. $x^4 - 2x + 4$
3. $7x^3 - 4x + 70$

Proof. 1. $2x - 1$ is of degree 1 so it is irreducible in $\mathbb{Q}[x]$. In addition, it is primitive.

2. It is primitive. So it is irreducible in $\mathbb{Z}[x]$ iff it is irreducible in $\mathbb{Q}[x]$. It has no rational roots by the rational root test. So it can only factor into two quadratic factors. Say $x^4 - 2x + 4 = (x^2 + ax + b)(x^2 + cx + d)$. Here we assume both factor has leading coefficients 1. Since there is no cubic term, we know $(a + c)x^3 = 0$ which implies $c = -a$. Now we expand $(x^2 + ax + b)(x^2 - ax + d) = x^4 + (b + d - a^2)x^2 + a(d - b)x + bd$. This implies $b + d - a^2 = 0$, $a(b - d) = -2$, and $bd = 4$. This system of equation has no real solutions.

3. Apply Eisenstein criterion for $p = 2$. □

Problem 3. For any complex number $w = a + bi \in \mathbb{C}$, its complex conjugate is $\bar{w} = a - bi \in \mathbb{C}$. For any polynomial $f(x) = \sum_{i=0}^n a_i x^i$ in $\mathbb{C}[x]$, its complex conjugate is $\overline{f(x)} = \sum_{i=0}^n \bar{a}_i x^i$ in $\mathbb{C}[x]$.

1. Verify that $\overline{a + b} = \bar{a} + \bar{b}$ and $\overline{ab} = \bar{a}\bar{b}$ for $a, b \in \mathbb{C}$.
2. Verify $\overline{f(x)g(x)} = \overline{f(x)}\overline{g(x)}$.
3. Suppose $w \in \mathbb{C}$ is a root of $f(x) \in \mathbb{C}[x]$. Then $\bar{w}$ is a root of $\overline{f(x)}$.
4. Suppose $w \in \mathbb{C}$ is a root of $f(x) \in \mathbb{R}[x]$. Then $\bar{w}$ is a root of $f(x)$.
5. For any $w \in \mathbb{C}$, $(x - w)(x - \bar{w})$ is in $\mathbb{R}[x]$.
6. Suppose $w \in \mathbb{C}$ is a root of $f(x) \in \mathbb{R}[x]$. Then $(x - w)(x - \bar{w})$ divides $f(x)$ in $\mathbb{R}[x]$.
7. Assume every polynomial in $\mathbb{C}[x]$ has a root in $\mathbb{C}$. Show any irreducible polynomial in $\mathbb{R}[x]$ must be of degree one or two.
8. Assume every polynomial in $\mathbb{C}[x]$ has a root in $\mathbb{C}$. Show any irreducible polynomial in $\mathbb{C}[x]$ must be of degree one.

Problem 4. Let R be a commutative ring and $p \in R$. Define the evaluation map $ev_p : R[x] \rightarrow R$ by $f(x) \mapsto f(p)$, which is a ring homomorphism. We say p is a root of $f(x) \in R[x]$ if $ev_p(f(x)) = f(p) = 0$ in R .

1. Show p is a root of $f(x)$ iff $f(x) = (x - p)g(x)$ for some $g(x) \in R[x]$.
2. What is the kernel of ev_p in $R[x]$?
3. Show that when R is a domain, then $(x - p)$ is a prime ideal in $R[x]$.

proof of part 1. ($\implies$) Suppose p is a root and write $f(x) = \sum_{i=0}^n a_i x^i$. Then $f(p) = 0$ and $f(x) - f(p) = f(x)$. Note we can write the LRS as

$$f(x) - f(p) = \sum_{i=0}^n a_i (x^i - p^i)$$

and $x^n - p^n = (x - p)(x^{n-1} + x^{n-2}p + \cdots + p^{n-1})$. So $x - p$ divides each term in $f(x) - f(p)$.

($\impliedby$) This direction is easy because evaluation map is a ring homomorphism. □

proof of part 3. Suppose $fg \in (x - p)$. Then $f(p)g(p) = 0$ in R which is domain iff $f(p) = 0$ or $g(p) = 0$ iff $f(p) \in \ker ev_p$ or $g(p) \in \ker ev_p$. □